8.1.1 bdh

(b) 1, 1, 2, 3, 5, 8, 13, 21, 34, 55

(d) 1, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, 1/9, 1/10

(h) 1, 49, 16, 25, 36, 49, 64, 81, 100

8.1.2 bd

(b) 2, 5, 8, 11, 14, 17

(d) 3, 5/2, 2, 3/2, 1, 1/2

8.1.3 bd

(b) 2, 5, 8, 11, 14, 17

(d)

8.1.4 bd

(b) Initial Value Geometric Sequence is a Common

Ratio of geometric sequence is r The general form of geometric sequence is $\{a, ar, ar^2, ar^3, \ldots\}$

(d) the initial value $a \in \mathbb{R}$ and common difference $d < 0$ For all cases

8.2.1. bF

(b) $\sum_{K=0}^{4} 2^K = 2^0 + 2^1 + 2^2 + 2^3 + 2^4$

$= 1 + 2 + 4 + 8 + 16$

$= 31$

(F)

8.2.2 b.d

(b) $\sum_{n=-2}^{4} n$

(d) $\sum_{i=0}^{9} (n^3)$

8.2.4 b

$J = K - 1$

$$\sum_{K=0}^{n-1} 2^{K-2} \rightarrow \sum_{J+1=0}^{n-1} 2^{(J+1)-2}$$

$$\sum_{J=-1}^{n-2} 2^{J-1} \rightarrow \text{Answer.}$$

8.2.6 b

$50 \times (23 \times 24 + 3 \times 23 \times 12)$

8.2.7 b

$P_n = (1.12) P_{n-1}$ where $P_0 = 30$

8.3.1 bdf

third Term (a_3)

$a_3 = 2a_2 + 3a_1$

$= 2 \times 5 + 3 \times 1$

$10 + 3$

$q_3 = 13$

Forth Term (q_4)

$q_4 = 2q_3 + 3q_2$

$= 2 \times 13 + 3 \times 5$

$= 26 + 15$

$q_4 = 41$

fifth Term (q_5)

$q_5 = 2q_4 + 3a_3$

$q_5 = 2 \times 41 + 3 \times 13$

$82 + 39$

$q_5 = 121$

Sixth Term (q_6)

$q_6 = 2q_5 + 3q_4$

$= 2 \times 121 + 3 \times 41$

$= 242 + 123$

$q_6 = 365$

Hence, the First Six Term of given Sequences are,
1, 5, 13, 41, 121, 365

(d) $C_1 = 4$

$C_2 = 5$

$C_3 = C_2 \cdot C_1 = 5 \times 4 = 20$

$C_4 = C_3 \cdot C_2 = 20 \times 5 = 100$

$C_5 = C_4 \cdot C_3 = 100 \times 20 = 2000$

$C_6 = C_5 \cdot C_4 = 2000 \times 100 = 200000$

(f) $f_1 = 0$

$f_3 = 5 \times 2 - 2 \times 0 = 10$

$f_4 = 5 \times 10 - 2 \times 2 = 46$

$f_5 = 5 \times 46 - 2 \times 10 = 210$

$f_6 = 5 \times 210 - 2 \times 46 = 958$

8.3.2 b

(b) Let P be the Required monthly payment,
Then the Condition is $a_n < a_{n-1}$

$$a_{n-1}\left(1+\frac{i}{100}\right) - P \leq a_{n-1}$$

$$a_{n-1}\left(1+\frac{i}{100} - 1\right) \leq P$$

$$a_{n-1} \leq P\left(\frac{100}{0.5}\right)$$

Here $n = 1$ must be true

$a_{i+1} \leq P\left(\frac{100}{0.5}\right)$

$a_0 \leq P(200)$

$30000 \leq P(200)$

$P \geq \frac{30000}{200}$

$P \geq 150$

$150

8.4.1 bdf

(b) $P(K) := \sum_{J=1}^{K} J^2 = \frac{K(K+1)(2K+1)}{6}$

(d) $\sum_{J=1}^{1} J^2 = 1^2 = 1$

R.H.S

$\sum^{K+1} J^2 = 1(1+1)(2+1)$

$j=1$ $\frac{\qquad\qquad}{6}$

(F) $\sum_{j=1}^{k} j^2 = \frac{k(k+1)(2k+1)}{6}$.